Linear Algebra: Review

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Linear Algebra: Review
(Vector Spaces)

Learning Objectives

- In this section, we will study the concepts related to the vector spaces. In particular, we will study
  - Group
  - Field
  - Vector space
  - Metric
  - Norm
  - Inner product
  - Basis vectors
  - Linear independence of basis vectors
  - Span
  - Basis

Notations

- Scalars will be written as small case letters in italics: x
- Vectors will be written as small case letters in bold: $\mathbf{x}$
- Matrices will be written as upper case letters in bold: $\mathbf{X}$
- The $(i,k)^{\text{th}}$ component of a matrix $\mathbf{X}$ will be written as X_{ik}
- Single random variable will be written as upper case letter in italics: X
- Vector random variable will be written as upper case, italics, and bold: $\boldsymbol{X}$
- Complex conjugation of a variable x will be denoted as: $\bar{x}$
- Transpose of a vector will be represented using superscript T: $\mathbf{x}^{\mathrm{T}}$
- Complex conjugate transpose of a vector will be represented using superscript H: $\mathbf{x}^{\mathrm{H}}$

Group

A group $(G, *)$ consists of the following:

(i) A set G

(ii) A rule or binary operation '$*$' (set G is closed under this operation) which associates with each pair of elements x and y in G, an element $(x * y)$ in G such that

a. This binary operation is associative:

i.e., $x*(y*z)=(x*y)*z \qquad \forall\, x, y, z \in G.$

b. There exists an element 'e' called identity of group G s.t. $e*x=x*e=x \quad \forall\, x \in G$

c. For every $x \in G$, there exists an element x^{-1} such that

$x*x^{-1}=e=x^{-1}*x \qquad \forall\, x \in G$

d. If $x*y=y*x \quad \forall\, x, y \in G$

⟹ Commutative group or Abelian group

Examples

1. Set of n x n invertible matrices with matrix multiplication
2. Set of continuous time periodic signals with time period T under binary operation '$*$' = '+'

Field

A field $(F, +, \cdot)$ consists of the following:

(i) A set F

(ii) Two binary operations '+' and '$\cdot$' such that

a. $(F, +)$ is an Abelian group

b. Define $F^*=F-\{0\}$. $(F^*, \cdot)$ is an Abelian group

c. The multiplication operation distributes over addition:

- Left distributive:

$$x\cdot(y+z)=x\cdot y + x\cdot z \qquad \forall\, x, y, z \in F$$

- Right distributive:

$$(x+y)\cdot z=x\cdot z + y\cdot z \qquad \forall\, x, y, z \in F$$

Examples: Check for F = R; F = C; F= Z; F=Q

Vector Space

A set V together with a map

$$\text{'+'}: V \times V \rightarrow V$$

$$(\mathbf{v}_1, \mathbf{v}_2) \rightarrow (\mathbf{v}_1 + \mathbf{v}_2) \text{ called vector addition}$$

And $\text{'.'}: F \times V \rightarrow V$

$$(\alpha, \mathbf{v}) \rightarrow (\alpha \cdot \mathbf{v}) \text{ called scalar multiplication}$$

is called a F-vector space or vector space over the field F if the following are satisfied:

(i) $(V, +)$ is an Abelian group

(ii) $\alpha \cdot (\mathbf{v}_1 + \mathbf{v}_2) = \alpha \cdot \mathbf{v}_1 + \alpha \cdot \mathbf{v}_2$ $\quad \forall\ \mathbf{v}_1, \mathbf{v}_2 \in V$ and $\forall\ \alpha \in F$

(iii) $(\alpha_1 + \alpha_2) \cdot \mathbf{v} = \alpha_1 \cdot \mathbf{v} + \alpha_2 \cdot \mathbf{v}$ $\quad \forall\ \mathbf{v} \in V$ and $\forall\ \alpha_1, \alpha_2 \in F$

(iv) $(\alpha\beta) \cdot \mathbf{v} = \alpha(\beta\mathbf{v}) = \beta(\alpha\mathbf{v})$ $\quad \forall\ \mathbf{v} \in V$ and $\forall\ \alpha, \beta \in F$

(v) $1 \cdot \mathbf{v} = \mathbf{v}$ $\quad \forall\ \mathbf{v} \in V$

(vi) $\alpha \cdot \mathbf{0} = \mathbf{0}$ $\quad \forall\ \alpha \in F$

(vii) $0 \cdot \mathbf{v} = \mathbf{0}$ $\quad \forall\ \mathbf{v} \in V$

(viii) If $\mathbf{v} \neq \mathbf{0}$, then $\alpha \cdot \mathbf{v} = \mathbf{0}$ implies that $\alpha = 0$.

(ix) If V is a vector space over a field F, then any linear combination of vectors lying in V (with scalars from F) would again lie in V.

Vector Space

Examples:

1. Set of continuous time periodic signals with the same time period T over the field $F = R$ form a vector space
2. Set of finite energy signals over the field $F = R$ form a vector space
3. $V_n(F)$ = n-dimensional vector space over the field $F = R$ or C

Metric Space

Metric is a map

$$d: X \times X \rightarrow R$$

that satisfies the following:

(i)	$d(x, y) \geq 0$ and $d(x, y)=0$	iff $x = y$	$\forall\, x, y \in X$
(ii)	$d(x, y) = d(y, x)$		$\forall\, x, y \in X$
(iii)	$d(x, y) \leq d(x, z) + d(z, y)$		$\forall\, x, y, z \in X$

This map is called a **metric** and a set equipped with this map is called a **metric space** and is denoted by (X, d).

Note that metric is a generalization of the notion of distance. An example is Euclidean distance.

Norm

Let V be a F-vector space. A map

$\|\cdot\|: V \to R$ is called a **norm** if it satisfies the following:

(i) $\|\mathbf{v}\| \geq 0$ and $\|\mathbf{v}\| = 0$ iff $\mathbf{v}=0$ $\quad \forall\, \mathbf{v} \in V$

(ii) $\|\alpha\mathbf{v}\| = |\alpha|\,\|\mathbf{v}\|$ $\quad \forall\, \mathbf{v} \in V$ and $\forall\, \alpha \in F$

(iii) $\|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\|$ $\quad \forall\, \mathbf{v}_1, \mathbf{v}_2 \in V$

A vector space equipped with a norm is called a **normed vector space**.

Example: Let V be a F-vector space equipped with a norm.

Prove that $d(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\|$ is a proper metric.

Inner Product

Let V be a F-vector space.

A map

$$< , > : V \times V \rightarrow F$$

is called an inner product if it satisfies the following:

(i) $< \mathbf{v}, \mathbf{v} > \geq 0$ and $< \mathbf{v}, \mathbf{v} > = 0$ iff $\mathbf{v} = 0$ $\quad \forall\ \mathbf{v} \in V$

(ii) $< \mathbf{v}_1, \mathbf{v}_2 > = \overline{< \mathbf{v}_2, \mathbf{v}_1 >}$ $\quad \forall\ \mathbf{v}_1, \mathbf{v}_2 \in V$

'$\bar{}$' denotes the complex conjugate operation.

(iii) It is linear in the first coordinate.

$< \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2, \mathbf{w} > = \alpha_1 < \mathbf{v}_1, \mathbf{w} > + \alpha_2 < \mathbf{v}_2, \mathbf{w} >$ $\forall\ \mathbf{v}_1, \mathbf{v}_2, \mathbf{w} \in V$ and $\forall\ \alpha_1, \alpha_2 \in F$

(iv) It is conjugate linear in the second coordinate.

$< \mathbf{v}, \alpha_1 \mathbf{w}_1 + \alpha_2 \mathbf{w}_2 > = \overline{\alpha_1} < \mathbf{v}, \mathbf{w}_1 > + \overline{\alpha_2} < \mathbf{v}, \mathbf{w}_2 >$

$\forall\ \mathbf{w}_1, \mathbf{w}_2, \mathbf{v} \in V$ and $\forall\ \alpha_1, \alpha_2 \in F$

Inner Product

Example 1: Define $\|\mathbf{v}\| = <\mathbf{v}, \mathbf{v}>^{1/2}$. Show that it is a proper norm.

Thus, if a space is an inner product space, we can define norm and then define metric.

Example 2: Consider $V_n(F)$, where vectors are n-tuples of scalars, i.e., $\mathbf{x} \in V$ and $x_i \in F$

i.e.,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ x_n \end{bmatrix}$$

Define inner product as: $<\mathbf{x}, \mathbf{y}> = \sum_{i=1}^{n} x_i \bar{y}_i$

Show that it satisfies all the properties of an inner product.

Linear Independence of Vectors

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ in $V_n(F)$ is called linearly independent (LI) if $\sum_{i=1}^{n} \alpha_i \mathbf{v}_i = 0$ implies $\alpha_i = 0$ for all i.

Definition: The number of maximal LI vectors in a vector space V is called the **dimension** of the vector space and the maximal LI vectors is called a **basis** for V.

If $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is a basis for $V_n(F)$, then for any $\mathbf{v} \in V$ and for some $\alpha_i \in F$

$$\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{e}_i$$

Orthogonal and Orthonormal Basis

Definition: A set of basis vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spanning an inner product space V is called **orthogonal** basis if

(i) $\mathbf{v}_i \neq 0 \qquad \forall \; i$

(ii) $< \mathbf{v}_i, \mathbf{v}_j > = 0 \qquad \forall \; i \neq j$

If additionally, the below condition is satisfied:

(iii) $< \mathbf{v}_i, \mathbf{v}_j > = 1 \qquad \forall \; i = j$

Then, the set of basis vectors is called as the orthonormal basis. In this case, we can also write the above conditions as $< \mathbf{v}_i , \mathbf{v}_j > = \delta_{ij}$.

Summary

- In this section, we studied the concepts of
  - Group
  - Field
  - Vector space
  - Metric
  - Norm
  - Inner product
  - Basis vectors
  - Linear independence of basis vectors
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with examples.

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